Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiments 8, 9

Effect of Attention Type and Attention Position in a Pretrained ResNet50

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
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1. Objective

The objective of this experiment is to understand how different attention mechanisms modify a pretrained convolutional neural network and how the **position at which attention is inserted** influences image classification performance.

A single pretrained architecture,

ResNet50 pretrained on ImageNet

is used throughout the experiment.

Students will investigate:

- channel attention;
- spatial attention;
- sequential channel–spatial attention;
- lightweight self-attention;
- different positions for inserting attention within ResNet50;
- single-stage and multi-stage attention;
- attention ordering;
- additional established attention mechanisms; and
- qualitative interpretation using internal attention maps and Grad-CAM.

The experiment emphasizes that the effectiveness of attention depends not only on the attention mechanism itself, but also on the characteristics of the feature representation on which it operates.

2. Learning Outcomes

After completing this experiment, students will be able to:

1. explain the stage-wise architecture of a pretrained ResNet50;
2. distinguish early, intermediate, and late convolutional features;
3. explain the motivation behind attention mechanisms;
4. implement channel attention;
5. implement spatial attention;
6. implement sequential channel–spatial attention;
7. implement a lightweight self-attention mechanism;
8. insert an attention module at different stages of a pretrained network;
9. analyze the influence of attention position;
10. compare different attention mechanisms under a controlled experimental protocol;
11. analyze classification accuracy and class-wise performance;
12. visualize internal attention maps;
13. generate Grad-CAM explanations;
14. distinguish an internal attention map from a post-hoc explanation map;
15. analyze accuracy–complexity trade-offs; and
16. perform controlled ablation studies.

3. Dataset

Recommended Dataset: TensorFlow Flowers (`tf_flowers`)

The TensorFlow Flowers dataset contains natural RGB flower images belonging to five classes:

`daisy`, `dandelion`, `roses`, `sunflowers`, `tulips`.

The images contain variations in:

- scale;
- viewpoint;
- illumination;
- background clutter;
- flower size;
- color;
- texture; and
- object location.

These variations make the dataset suitable for studying the behavior of attention mechanisms.

3.1 Image Preprocessing

Resize every input image to

$$224 \times 224 \times 3.$$

The image should be preprocessed using the preprocessing procedure expected by the ImageNet-pretrained ResNet50.
Suitable data augmentation may include:

- random horizontal flipping;
- small rotations;
- random zoom;
- random translation; and
- mild contrast variation.

The same augmentation strategy must be maintained for all model comparisons.

3.2 Dataset Split

Use the same training, validation, and testing split for every experiment:

$$\boxed{70\% : 15\% : 15\%}$$

for

$$\text{Training} : \text{Validation} : \text{Testing}.$$

The exact same samples should be retained in each subset for all ablation experiments.
This is necessary to ensure that differences in performance are caused by changes in the model rather than changes in the dataset split.

4. Pretrained Architecture: ResNet50

ResNet50 pretrained using ImageNet is used as the **only backbone architecture** in this experiment.
For an input image

$$I \in \mathbb{R}^{224\times224\times3},$$

the major ResNet50 stages produce approximately the following feature dimensions:

ResNet50 Stage	**Spatial Resolution**	**Channels**
Stem + MaxPool	56×56	64
conv2_x	56×56	256
conv3_x	28×28	512
conv4_x	14×14	1024
conv5_x	7×7	2048

As the network becomes deeper:

$$\boxed{\text{Spatial Resolution} \downarrow}$$

while

$$\boxed{\text{Feature Channels} \uparrow}$$

and the representation generally progresses from low-level visual information toward higher-level semantic information.

5. Understanding ResNet50 Feature Hierarchy

The approximate interpretation of different stages is:

- **Early features:** edges, colors, orientations, boundaries, and simple textures.
- **Intermediate features:** local patterns, repeated structures, flower parts, petal arrangements, and shape information.
- **Late features:** high-level object structures and class-discriminative semantic information.

Therefore, the same attention mechanism may produce different behavior when inserted at different locations.

6. Why Attention Position Matters

Attention is not independent of the features on which it operates.
Suppose an attention module receives

$$F \in \mathbb{R}^{H \times W \times C}.$$

The values of

$$H, \qquad W, \qquad C$$

and the semantic meaning of the feature tensor change significantly with network depth.
Therefore:

- **early-stage attention** operates on high-resolution, low-level visual information;
- **middle-stage attention** operates on intermediate features containing both spatial and semantic information; and
- **late-stage attention** operates on low-resolution, high-level semantic representations.

Thus, students should investigate both:

$$\boxed{\text{Which attention mechanism should be used?}}$$

and

$$\boxed{\text{Where should the attention mechanism be inserted?}}$$

7. Candidate Attention Positions

Define four candidate attention positions in ResNet50.

$$P_1: \text{ after conv2_x},$$

$$P_2: \text{ after conv3_x},$$

$$P_3: \text{ after conv4_x},$$

$$P_4: \text{ after conv5_x}.$$

The corresponding feature dimensions are:

Position	Location	Resolution	Channels
P_1	After conv2_x	56×56	256
P_2	After conv3_x	28×28	512
P_3	After conv4_x	14×14	1024
P_4	After conv5_x	7×7	2048

For the principal laboratory exercise, students should concentrate on

$$\boxed{P_2, \quad P_3, \quad P_4}$$

because P_1 contains comparatively high-resolution features and may substantially increase the computational cost of some attention mechanisms.

8. Baseline Model

The baseline is the pretrained ResNet50 without any additional attention mechanism.



The baseline establishes the reference performance against which the attention-based models are compared.

9. Channel Attention

Consider a feature tensor

$$F \in \mathbb{R}^{H \times W \times C}.$$

Channel attention determines the relative importance of different feature channels.
Global average pooling is first used to generate a channel descriptor:

$$z_c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} F(i, j, c).$$

Thus,

$$z \in \mathbb{R}^C.$$

The channel descriptor is transformed using two learnable transformations:

$$M_c(F) = \sigma\left(W_2 \delta\left(W_1 z\right)\right),$$

where:

- $\delta(\cdot)$ denotes a nonlinear activation such as ReLU;
- $\sigma(\cdot)$ denotes the sigmoid activation;
- W_1 reduces the channel representation; and
- W_2 restores the channel dimension.

The refined feature tensor is:

$$F_c = F \odot M_c(F).$$

Channel attention attempts to answer:

$$\boxed{\text{Which feature channels are important?}}$$

A reduction ratio of

$$r = 16$$

may be used as the default setting throughout the laboratory experiment.
The reduction ratio need not be treated as a separate mandatory ablation.

10. Spatial Attention

Spatial attention determines the spatial regions that should receive greater importance.
For the input feature tensor

$$F \in \mathbb{R}^{H\times W\times C},$$

compute average pooling across channels:

$$F_{\text{avg}} = \text{Mean}_c(F),$$

and maximum pooling across channels:

$$F_{\max} = \text{Max}_c(F).$$

Both tensors have dimensions

$$H \times W \times 1.$$

Concatenate them:

$$F_s = [F_{\text{avg}} \| F_{\max}].$$

The spatial attention map is then computed as

$$M_s(F) = \sigma\left(\text{Conv}_{7\times 7}(F_s)\right).$$

Therefore,

$$M_s(F) \in \mathbb{R}^{H\times W\times 1}.$$

The spatially refined feature is

$$F'_s = F \odot M_s(F).$$

Spatial attention answers:

$$\boxed{\text{Which spatial locations are important?}}$$

11. Sequential Channel–Spatial Attention

Channel and spatial attention can be sequentially applied.
First,

$$F_c = M_c(F) \odot F.$$

Then,

$$F_{cs} = M_s(F_c) \odot F_c.$$

The overall structure becomes:

$$\boxed{F \to \text{Channel Attention} \to \text{Spatial Attention} \to F_{cs}}$$

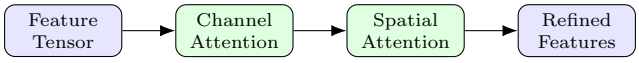
This arrangement first determines

"what is important"

and then determines

"where it is important."

A simplified diagram is shown below.



12. Self-Attention

Self-attention allows spatial locations to interact directly with other spatial locations.
For

$$F \in \mathbb{R}^{H \times W \times C},$$

flatten the spatial dimensions:

$$X \in \mathbb{R}^{N \times C},$$

where

$$N = HW.$$

Generate the query, key, and value matrices:

$$Q = XW_Q,$$

$$K = XW_K,$$

and

$$V = XW_V.$$

The attention matrix is

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right),$$

where d is the dimension of the query/key representation.
The attended output is

$$Y = AV.$$

After reshaping the output back to the original spatial structure, a residual formulation may be used:

$$F_{\text{SA}} = F + \gamma\,\text{reshape}(Y),$$

where γ is a learnable scaling parameter.
Self-attention attempts to answer:

Which spatial features are related to other spatial features?

Since the attention matrix has dimension approximately

$$N \times N,$$

its computational requirements increase rapidly as

$$N = HW$$

increases.
Therefore, self-attention should preferably be introduced at

$$P_3$$

or

$$P_4.$$

13. Inserting Attention into ResNet50

For any selected stage P_k, the network structure can be written as

$$\text{Previous ResNet Stage} \rightarrow \text{Attention} \rightarrow \text{Next ResNet Stage}.$$

For example, attention inserted after conv4_x gives:

$$\boxed{\text{conv4_x} \rightarrow \text{Attention} \rightarrow \text{conv5_x}}$$

For channel–spatial attention:

$$\boxed{\text{conv4_x} \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow \text{conv5_x}}$$

The student must record the feature dimensions immediately before and after the attention block.
Attention should not unintentionally modify the spatial or channel dimensions unless the selected mechanism explicitly requires such a transformation.

14. Main Experiment 1: Attention-Type Ablation

The first experiment studies the effect of different types of attention.
Fix the attention position at

$$\boxed{P_3 = \text{after conv4_x}}$$

for all models.
Compare the following:

Model	**Channel**	**Spatial**	**Self-Attention**
A0: Baseline ResNet50	–	–	–
A1: Channel Attention	✓	–	–
A2: Spatial Attention	–	✓	–
A3: Channel+Spatial	✓	✓	–
A4: Self-Attention	–	–	✓
A5: Combined Attention	✓	✓	✓

This experiment investigates:

$$\boxed{\text{How does attention type affect model performance?}}$$

Students must keep the following identical:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;
- learning rate schedule;
- epochs;
- batch size;
- backbone;
- classifier;
- attention position; and
- evaluation procedure.

15. Main Experiment 2: Attention-Position Ablation

The second experiment investigates the influence of the position of attention.
Fix the attention type as:

$$\boxed{\text{Channel Attention} \rightarrow \text{Spatial Attention}}$$

and insert exactly the same module at different stages.

Experiment	Attention Position	Resolution	Channels
B1	After conv2_x	56×56	256
B2	After conv3_x	28×28	512
B3	After conv4_x	14×14	1024
B4	After conv5_x	7×7	2048

Students should study whether attention is more beneficial for:

- high-resolution low-level features;
- intermediate representations; or
- low-resolution semantic features.

The principal question is:

$$\boxed{\text{How does the location of attention influence performance?}}$$

16. Main Experiment 3: Multiple Attention Positions

Attention may also be inserted at more than one ResNet50 stage.
Compare:

$$\text{Attention only at } P_3,$$

$$\text{Attention only at } P_4,$$

and

$$\text{Attention at } P_3 + P_4.$$

As an extended experiment, students may evaluate:

$$P_2 + P_3 + P_4.$$

The purpose is to investigate whether applying attention repeatedly provides sufficient performance improvement to justify the additional:

- parameters;
- computation;
- memory;
- training time; and
- inference time.

Students should experimentally answer:

$$\boxed{\text{Is attention at more locations necessarily more effective?}}$$

17. Main Experiment 4: Attention Order

The sequence in which channel and spatial attention are applied may influence the resulting features.
Compare:

$$\boxed{\text{Channel} \rightarrow \text{Spatial}}$$

against

$$\boxed{\text{Spatial} \rightarrow \text{Channel}}$$

while maintaining the same insertion position.
For example, at P_3:

$$F \rightarrow M_c \rightarrow M_s$$

should be compared with

$$F \rightarrow M_s \rightarrow M_c.$$

Students should determine whether selecting informative feature channels before selecting spatial regions produces behavior different from selecting spatial regions first.

18. Main Experiment 5: Spatial Attention Kernel Size

The receptive field used to generate the spatial attention map may influence the result.
Evaluate:

$$k \in \{3, 5, 7\}.$$

Use the same attention position and training protocol.

Spatial Kernel	Accuracy	Macro F1	Inference Time
3×3			
5×5			
7×7			

Students should discuss whether a larger spatial kernel provides a useful increase in contextual information.

19. Additional Task: Comparison of Different Attention Mechanisms

As an additional exploration task, students should compare the attention mechanisms implemented in the laboratory with established attention approaches.
Students should select at least

$$\boxed{\text{two additional attention mechanisms}}$$

from the following:

- Squeeze-and-Excitation (SE) Attention;
- Efficient Channel Attention (ECA);
- Convolutional Block Attention Module (CBAM);
- Coordinate Attention; and
- lightweight Self-Attention.

The pretrained backbone must remain:

$$\boxed{\text{ResNet50}}$$

for all comparisons.
To ensure a fair comparison, additional attention mechanisms should preferably be inserted at the same position:

$$\boxed{P_3 = \text{after conv4_x}}$$

unless the student provides a technical justification for using another location.

19.1 Suggested Comparison

Attention	Accuracy	Precision	Recall	Macro F1	Parameters	Inference Time
No Attention						
Channel / SE						
Spatial Attention						
ECA						
CBAM						
Self-Attention						
Additional Mechanism						

19.2 Questions for Attention-Mechanism Comparison

Students should investigate:

1. How does classification accuracy vary across attention mechanisms?
2. How does macro F1-score vary across attention mechanisms?
3. How many parameters are introduced by each attention module?
4. How does each attention mechanism influence inference time?
5. Does attention improve the convergence behavior?
6. Does attention produce more focused feature localization?
7. Does an increase in attention complexity always improve classification performance?
8. Does a lightweight attention mechanism provide comparable performance with a computationally heavier mechanism?

The objective is to understand:

$$\boxed{\text{Performance} + \text{Complexity} + \text{Attention Behaviour}}$$

rather than considering only classification accuracy.

20. Transfer-Learning Strategy

Use ImageNet pretrained weights.

A recommended training strategy is:

1. Load ResNet50 with ImageNet pretrained weights.
2. Remove the original ImageNet classification layer.
3. Insert the selected attention module.
4. Add Global Average Pooling.
5. Add Dropout.
6. Add a Dense classification layer containing five output neurons.
7. Initially freeze the pretrained ResNet50 layers.
8. Train the newly introduced attention and classification layers.
9. Unfreeze the final ResNet50 stage or selected final stages.
10. Fine-tune the network using a smaller learning rate.

A suitable initial learning rate for the newly introduced layers is

$$10^{-3}.$$

During fine-tuning, a smaller learning rate such as

$$10^{-5} \quad \text{to} \quad 10^{-4}$$

may be used.

The exact values selected by the student should remain identical for all controlled comparisons.

21. Loss Function

For the five-class classification problem, use categorical cross-entropy.

For N samples:

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{5} y_{i,c}\log \widehat{y}_{i,c},$$

where:

$$y_{i,c}$$

is the ground-truth label and

$$\widehat{y}_{i,c}$$

is the predicted probability.

The same loss formulation must be used for every attention comparison.

22. Evaluation Metrics

Students must report the following quantitative metrics:

- classification accuracy;
- macro precision;
- macro recall;
- macro F1-score;
- number of trainable parameters;
- total number of parameters; and
- average inference time per image.

For class c,

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c},$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c},$$

and

$$F1_c = 2\frac{\text{Precision}_c \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}.$$

Macro F1 is:

$$\text{Macro F1} = \frac{1}{C}\sum_{c=1}^{C} F1_c,$$

where

$$C = 5.$$

23. Required Main Result Table

Students should summarize the major experiments using the following table.

Model	Position	Accuracy	Macro Precision	Macro Recall	Macro F1	Trainable Params	Total Params	Inference Time (s/image)
Baseline	–	0.9183	0.9165	0.9192	0.9175	10,245	23,597,957	0.0169
Channel Attention	P_3	0.9129	0.9138	0.9140	0.9118	142,405	23,730,117	0.0174
Spatial Attention	P_3	0.9183	0.9175	0.9194	0.9182	10,343	23,598,055	0.0174
Channel+Spatial	P_2	0.9093	0.9092	0.9106	0.9094	43,655	23,631,367	0.0175
Channel+Spatial	P_3	0.9238	0.9223	0.9235	0.9227	142,503	23,730,215	0.0175
Channel+Spatial	P_4	0.9220	0.9208	0.9235	0.9219	536,807	24,124,519	0.0176
Self-Attention	P_3	0.9365	0.9358	0.9361	0.9356	1,189,894	24,777,606	0.0176
Combined Attention	P_3	0.9129	0.9149	0.9100	0.9122	1,322,152	24,909,864	0.0181
Channel+Spatial	$P_3 + P_4$	0.9201	0.9193	0.9201	0.9197	669,065	24,256,777	0.0178

24. Graphical Performance Analysis

Students should not present only numerical tables.

Appropriate graphs must be generated to analyze model behavior.

At least three of the following graphical comparisons should be included in the laboratory report.

24.1 Training and Validation Accuracy

Plot

$$\text{Epoch} \quad \text{versus} \quad \text{Accuracy}$$

for:

- baseline ResNet50; and
- selected attention models.

Both training and validation accuracy should be shown.
Students should comment on whether attention:

- accelerates convergence;
- produces smoother learning;
- reduces the train–validation gap;
- leads to overfitting; or
- causes unstable training.

24.2 Training and Validation Loss

Plot

$$\text{Epoch} \quad \text{versus} \quad \mathcal{L}_{CE}.$$

The graph should include training and validation loss.
Students should compare the convergence behavior of the baseline and attention models.

24.3 Attention Mechanism versus Accuracy

Generate a bar graph:

$$x = \text{Attention Mechanism}$$

and

$$y = \text{Classification Accuracy}.$$

Possible entries are:

$$\text{Baseline, Channel, Spatial, Channel+Spatial, Self-Attention, SE, ECA, CBAM.}$$

24.4 Attention Mechanism versus Macro F1

Similarly, plot:

$$x = \text{Attention Mechanism}$$

and

$$y = \text{Macro F1-score}.$$

This comparison is useful when the class-wise performance is not uniform.

24.5 Accuracy versus Model Parameters

Create a scatter plot using:

$$x = \text{Total Model Parameters}$$

and

$$y = \text{Test Accuracy}.$$

Each point should represent one attention mechanism.
This allows the student to study the

$$\boxed{\text{Accuracy–Complexity Trade-off}}$$

instead of evaluating accuracy independently.

24.6 Macro F1 versus Inference Time

Plot:

$$x = \text{Inference Time per Image}$$

and

$$y = \text{Macro F1-score.}$$

Students should analyze whether additional computational cost produces a meaningful increase in classification performance.

25. Confusion Matrix Analysis

Generate a confusion matrix for:

1. baseline ResNet50; and
2. the selected attention-based ResNet50.

The confusion matrix should contain all five flower classes.

Students should analyze:

- which classes are most accurately classified;
- which classes are frequently confused;
- which class benefits most from attention;
- whether attention reduces a particular type of confusion; and
- whether the improved overall accuracy is distributed across classes.

26. Qualitative Analysis of Attention

Quantitative classification performance does not directly explain where the network obtains evidence for its prediction.

Students must therefore perform qualitative analysis.

The visual analysis should include:

- internal attention maps;
- attention overlays;
- intermediate feature maps; and
- Grad-CAM explanations.

Students should preferably use the same input image when comparing different models.

27. Grad-CAM Visualization

Gradient-weighted Class Activation Mapping (Grad-CAM) provides a class-specific visualization of the regions contributing to a prediction.

Let

$$A^k$$

represent feature map k from a selected convolutional layer.

For class score y^c, the importance coefficient is:

$$\alpha_k^c = \frac{1}{HW} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}.$$

The Grad-CAM map is:

$$L_{\text{Grad}-\text{CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k \right).$$

The map should be resized to the original image resolution and overlaid on the input.

Students should compare:

Input | Baseline Grad-CAM | Attention Model Grad-CAM

for the same image.

27.1 Grad-CAM Interpretation

Students should qualitatively determine whether the attention model:

- focuses more strongly on the flower;
- suppresses irrelevant background regions;
- concentrates on petals;
- concentrates on the central flower structure;
- attends to discriminative edges;
- distributes activation across the entire flower;
- incorrectly focuses on leaves;
- incorrectly focuses on background objects; or
- changes its focus for an incorrect prediction.

Students should avoid unsupported statements such as:

> "The Grad-CAM of Model A is better."

Instead, the observation should describe visible evidence.
For example:

> The baseline network activates both the flower and surrounding leaves, whereas the attention-based model produces stronger activation over the petal region and weaker activation over the background.

28. Internal Attention Maps

An important objective of this experiment is to distinguish between:

Internal Attention

and

Post-hoc Explanation

because they are not the same.
An attention module directly computes weights during the forward pass.
Grad-CAM, in contrast, is generated after prediction using gradients.

28.1 Channel-Attention Visualization

For channel attention:

$$M_c(F) = [m_1, m_2, \ldots, m_C].$$

Students should visualize the learned channel importance values using a bar graph.
Instead of displaying every channel when C is large, the student may show the top-K channels.
Recommended values are:

$$K = 10$$

or

$$K = 20.$$

Students should comment on whether:

- a small number of channels receive strong weights;
- many channels contribute similarly; or
- channel weighting changes substantially for different input images.

28.2 Spatial-Attention Visualization

For spatial attention:

$$M_s(F) \in \mathbb{R}^{H \times W}.$$

Resize the attention map to:

$$224 \times 224$$

and overlay it on the input image.
Display:

$$\boxed{\text{Input} \mid \text{Spatial Attention Map} \mid \text{Attention Overlay}}$$

Students should identify the image regions receiving high attention values.

28.3 Self-Attention Visualization

For self-attention, students may visualize one of the following:

- attention associated with a selected query location;
- mean attention across query positions;
- maximum attention across query positions; or
- an aggregated spatial self-attention response.

The visualization should be reshaped into a spatial form and, when appropriate, resized to the image resolution.
Students should analyze how distant regions interact within the feature map.

29. Attention Position Visualization

Use the same test image to visualize spatial attention at different ResNet50 stages.
Generate:

$$M_s^{P_2}, \qquad M_s^{P_3}, \qquad M_s^{P_4}.$$

Display the maps as:

$$\boxed{\text{Input} \mid M_s^{P_2} \mid M_s^{P_3} \mid M_s^{P_4}}$$

The objective is to visually study how attention changes as feature representations become deeper.
Possible observations may include:

- P_2 responding strongly to edges, colors, textures, and fine structures;
- P_3 responding to flower parts and intermediate object structures;
- P_4 responding to larger semantic flower regions.

These observations must be based on the student's experimental visualizations.

30. Qualitative Comparison Across Attention Mechanisms

Select at least:

$$\boxed{\text{3 correctly classified images}}$$

and

$$\boxed{\text{2 incorrectly classified images.}}$$

For every selected image, compare the behavior of multiple models.
A recommended arrangement is:

Input Image
↓
Baseline Grad-CAM
Channel-Attention Grad-CAM
Spatial-Attention Map
Channel+Spatial Grad-CAM
Self-Attention Response

Students may alternatively arrange the images horizontally in the report.

30.1 Qualitative Criteria

The student should discuss:

- foreground localization;
- background suppression;
- localization of petals;
- localization of the flower center;
- activation compactness;
- activation coverage;
- incorrectly attended regions;
- differences between correctly and incorrectly classified samples;
- differences between shallow and deep attention; and
- consistency of attention across different images.

31. Required Visualizations

The laboratory report must contain the following visualizations:

1. sample input images from all five classes;
2. training and validation accuracy curves;
3. training and validation loss curves;
4. confusion matrix;
5. feature map before attention;
6. channel-attention weights;
7. spatial-attention heatmap;
8. spatial-attention overlay;
9. self-attention response for the self-attention experiment;
10. feature map after attention;
11. baseline Grad-CAM;
12. Grad-CAM for at least one attention-based model;
13. attention maps at P_2, P_3, and P_4; and
14. at least one graphical comparison of the performances of different attention mechanisms.

1 Visualizations and Observations

The following figures present sample images, learning curves, classification results, attention visualizations, and model comparisons. Each figure is followed immediately by its own observation.

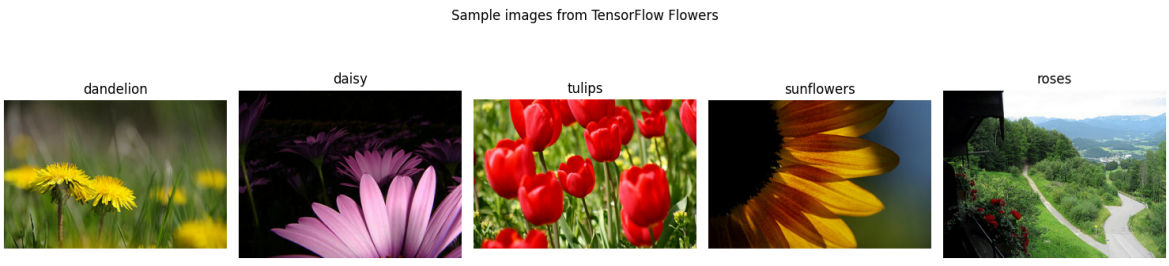


Figure 1: Sample input images from the five flower classes.

Observation: The dataset contains visually different examples of daisies, dandelions, roses, sunflowers, and tulips. These examples show the variation the classifier must learn.

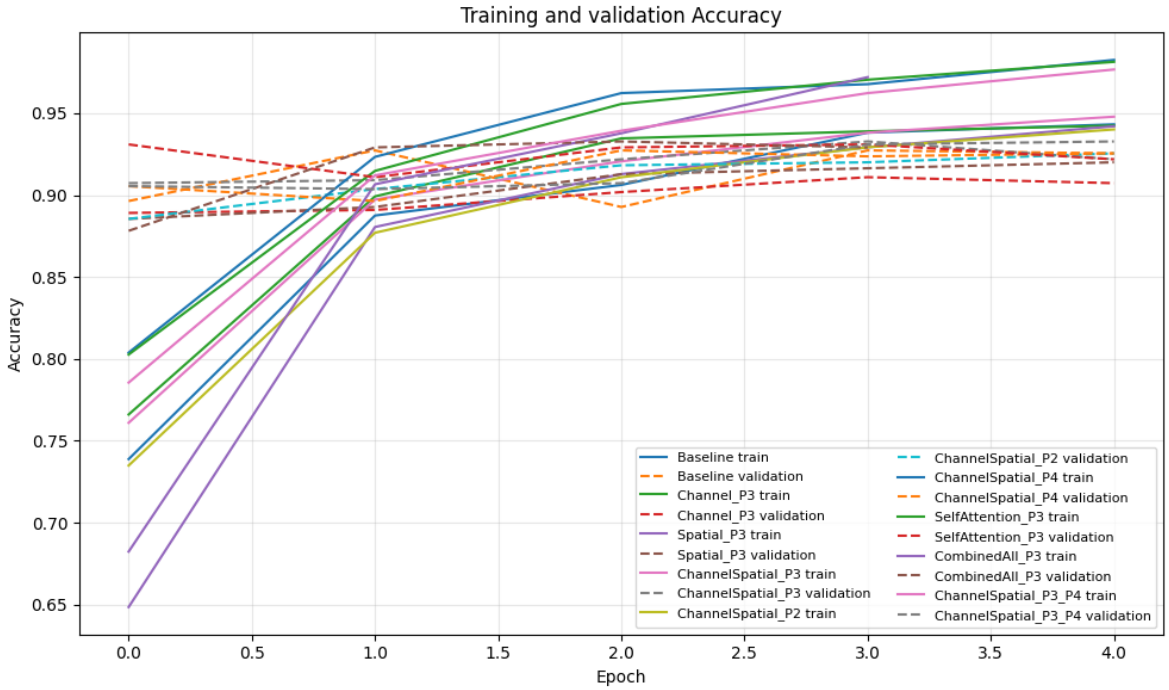


Figure 2: Training and validation accuracy curves.

Observation: Accuracy increases as the models learn useful image features. The validation curve helps assess generalization; a persistent gap between training and validation accuracy can indicate overfitting.

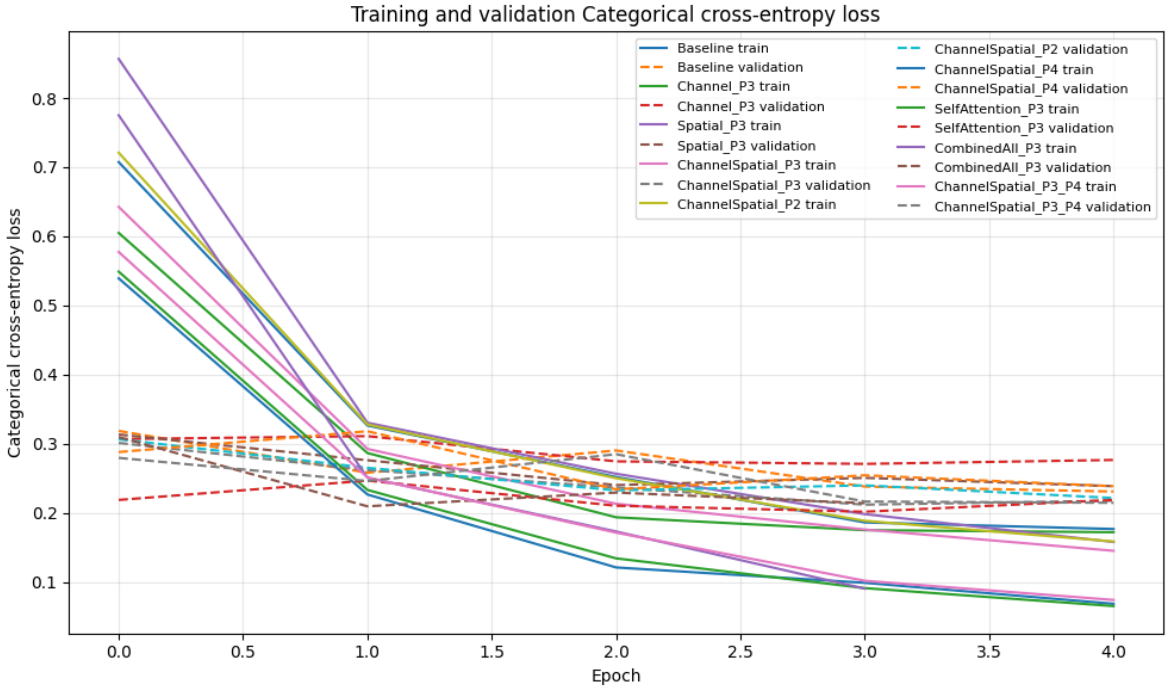


Figure 3: Training and validation loss curves.

Observation: The loss curves show how prediction error changes over epochs. Validation loss decreases overall for the baseline, reaching about 0.2309 in the final epoch.

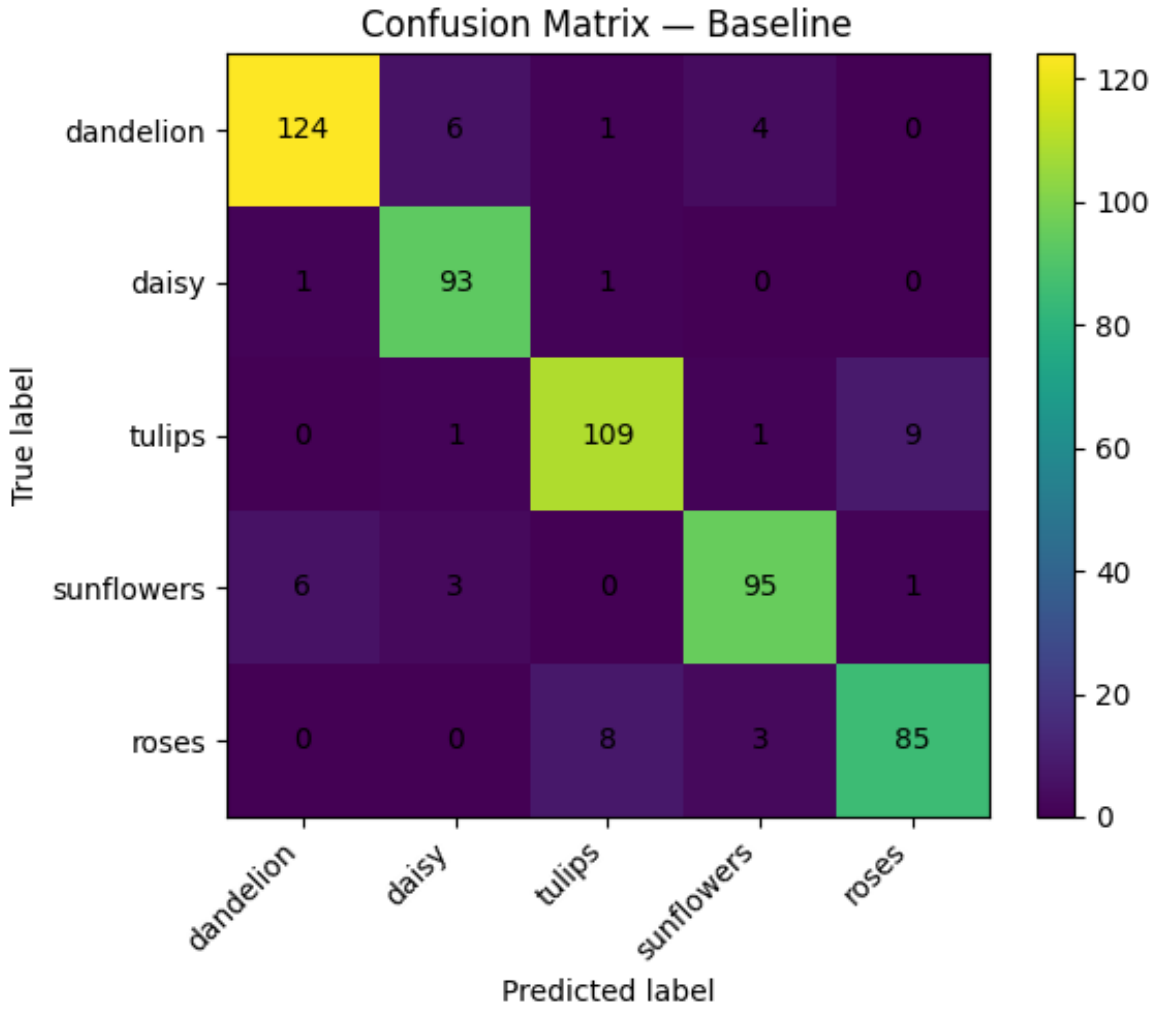


Figure 4: Confusion matrix for the baseline ResNet50 model.

Observation: The baseline achieves 91.83% test accuracy. Daisy has high recall (0.98), while roses have the lowest class-wise F1-score in the baseline report (0.89).

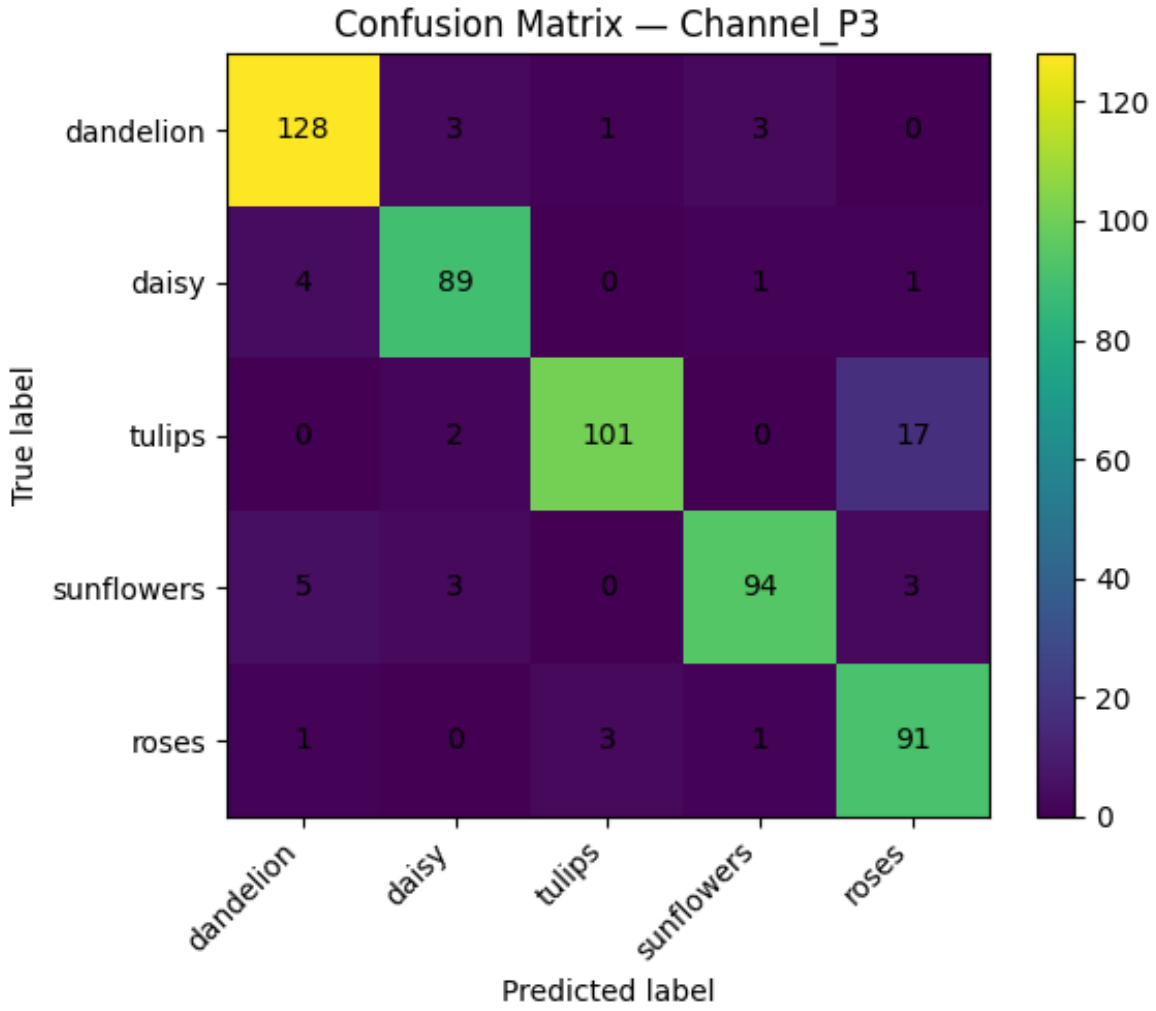


Figure 5: Confusion matrix for the Channel_P3 model.

Observation: Channel_P3 achieves 91.29% test accuracy, which is slightly below the baseline's 91.83%. Channel attention at P3 alone does not improve overall accuracy in this run.

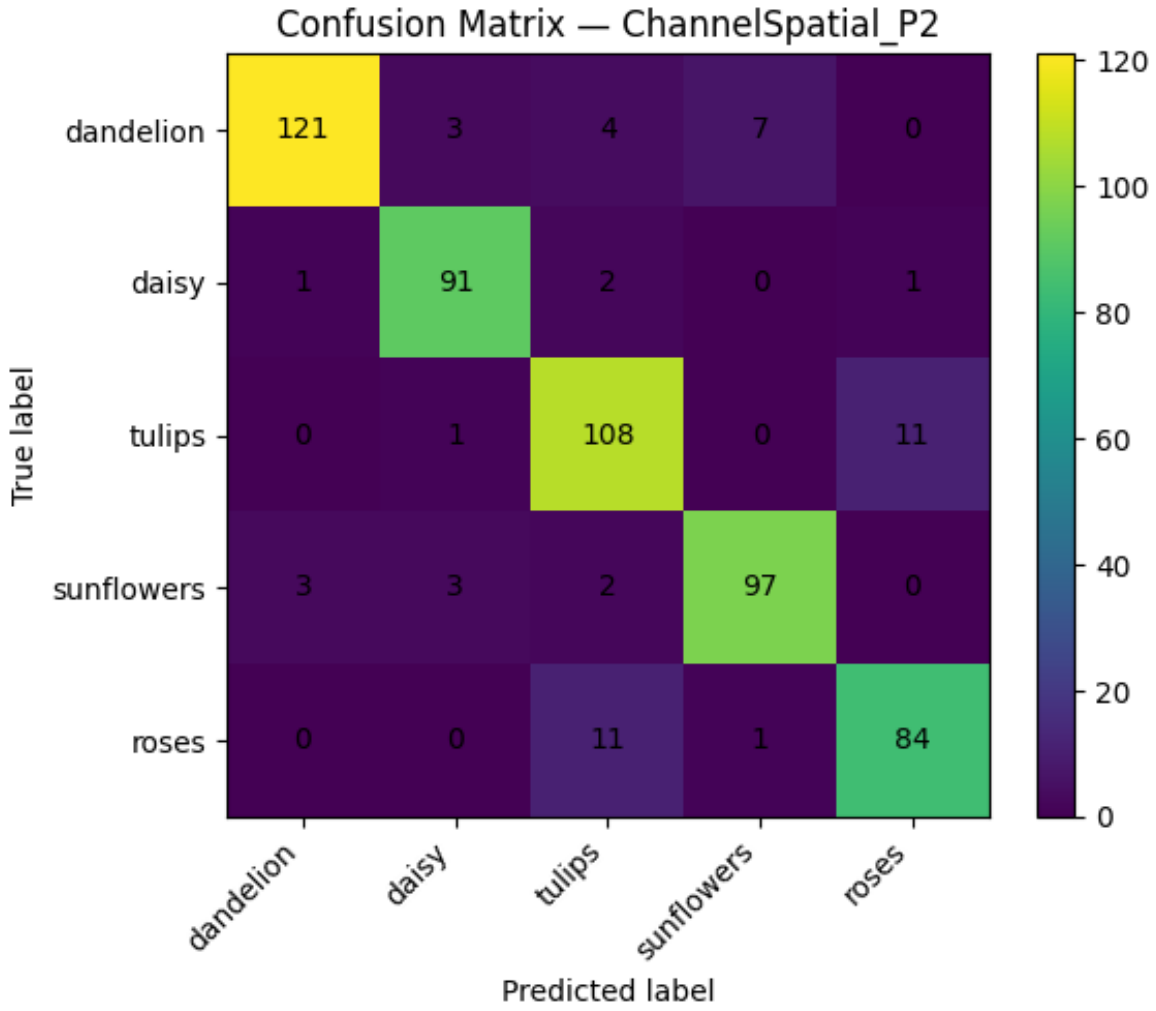


Figure 6: Confusion matrix for ChannelSpatial_P2.

Observation: ChannelSpatial_P2 reaches 90.93% test accuracy, the lowest among the nine evaluated configurations. In this experiment, placing the combined attention block at P2 was less effective than P3 or P4.

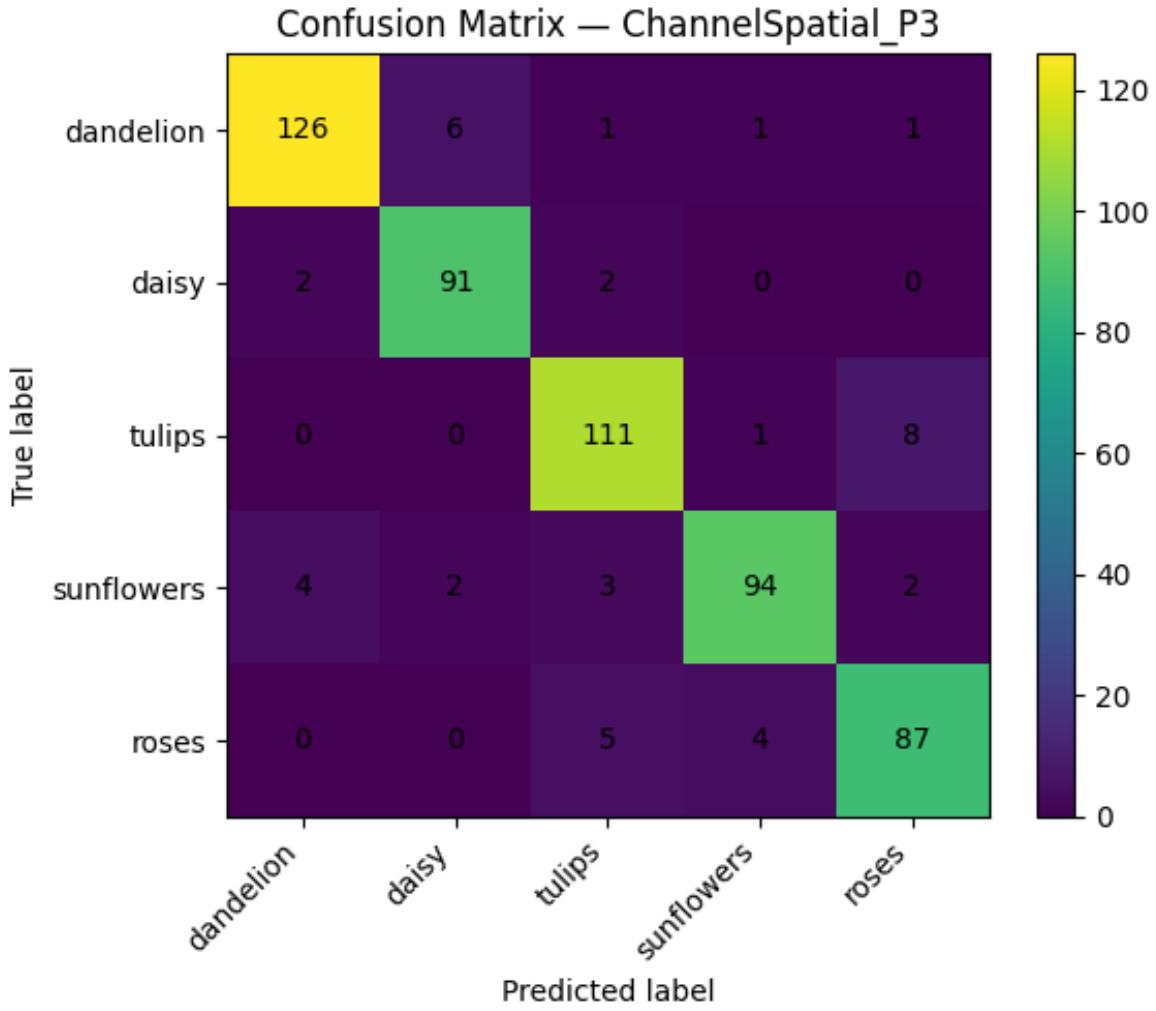


Figure 7: Confusion matrix for ChannelSpatial_P3.

Observation: ChannelSpatial_P3 achieves 92.38% test accuracy and a macro-F1 score of 0.9227. It improves on the baseline by 0.55 percentage points in test accuracy.

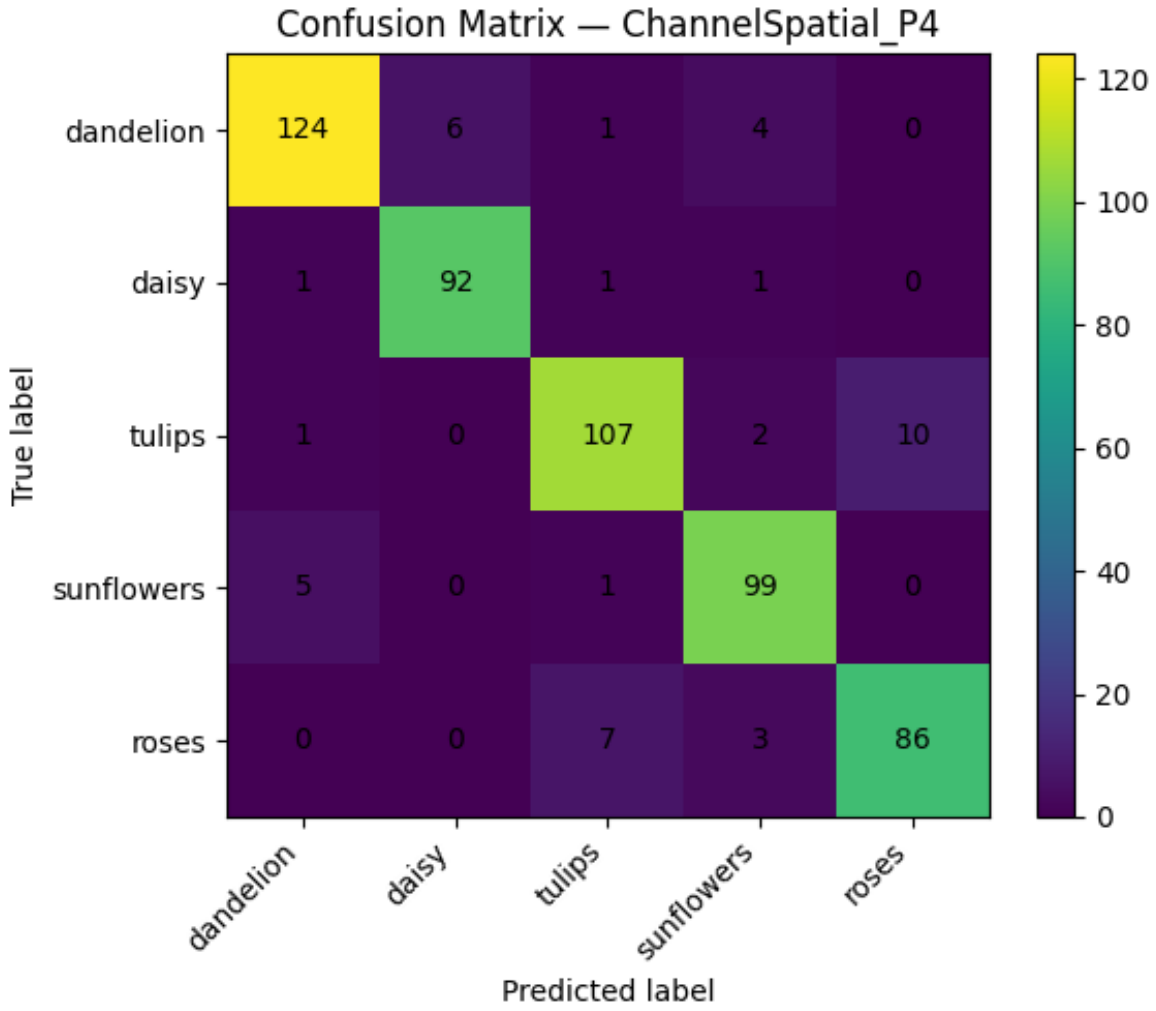


Figure 8: Confusion matrix for ChannelSpatial_P4.

Observation: ChannelSpatial_P4 reaches 92.20% test accuracy, slightly below ChannelSpatial_P3 at 92.38%. Both configurations outperform the baseline in this evaluation.

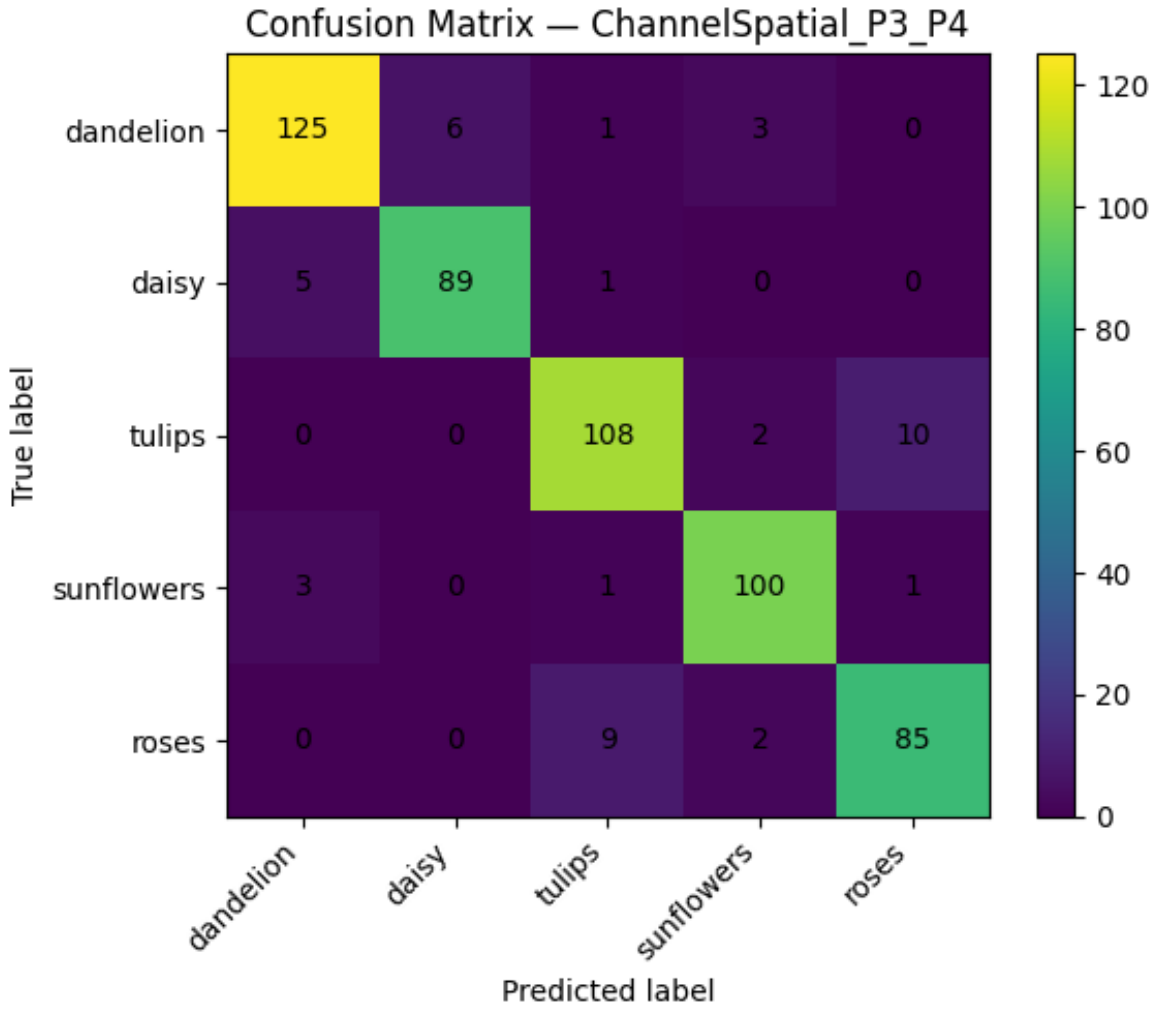


Figure 9: Confusion matrix for ChannelSpatial_P3_P4.

Observation: Applying channel-plus-spatial attention at both P3 and P4 gives 92.01% test accuracy. Using two stages does not outperform the single P3 placement in this run.

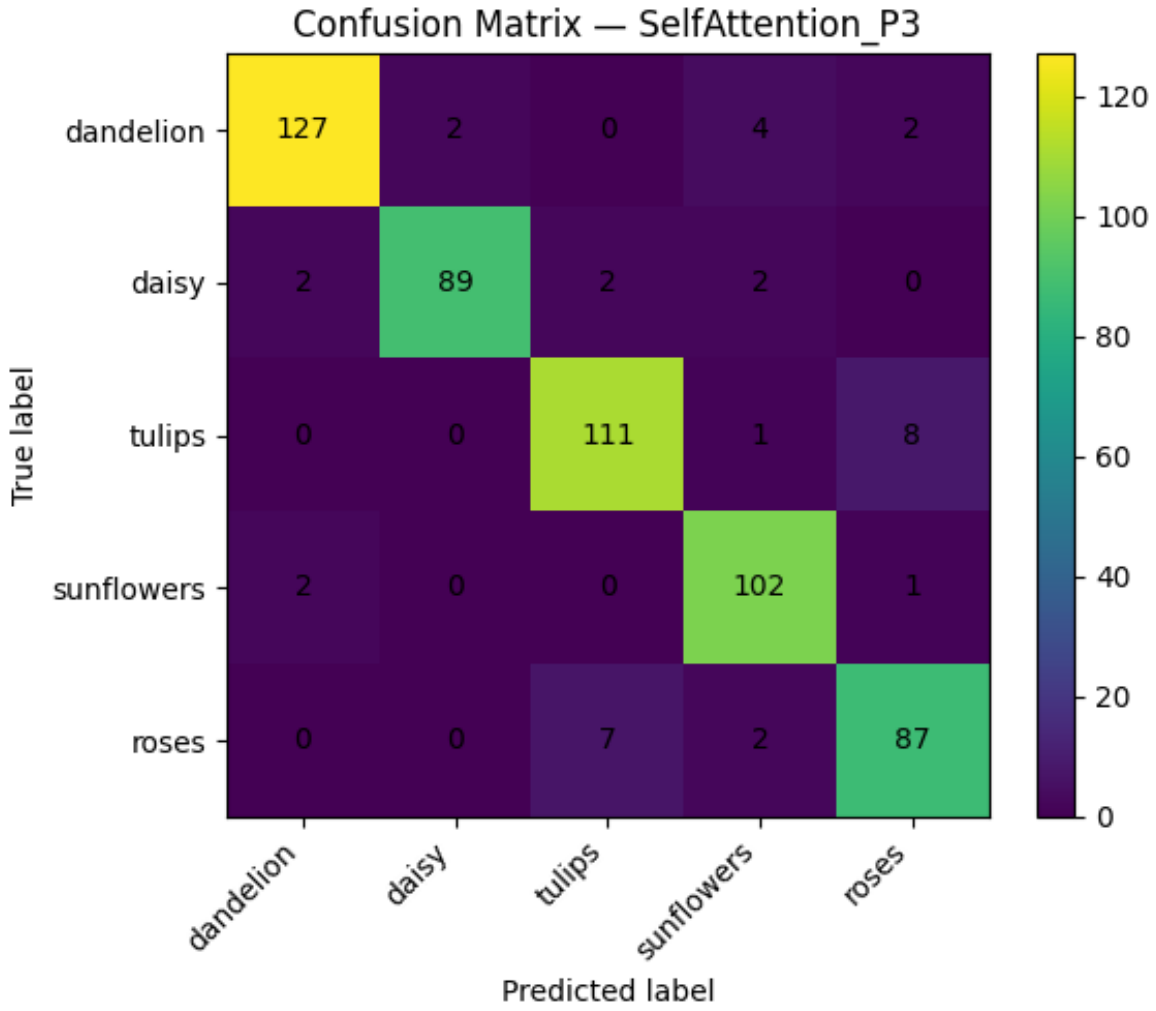


Figure 10: Confusion matrix for the Spatial_P3 model.

Observation: Spatial_P3 matches the baseline test accuracy at 91.83%. Its macro-F1 score is 0.9182, only slightly higher than the baseline's 0.9175.

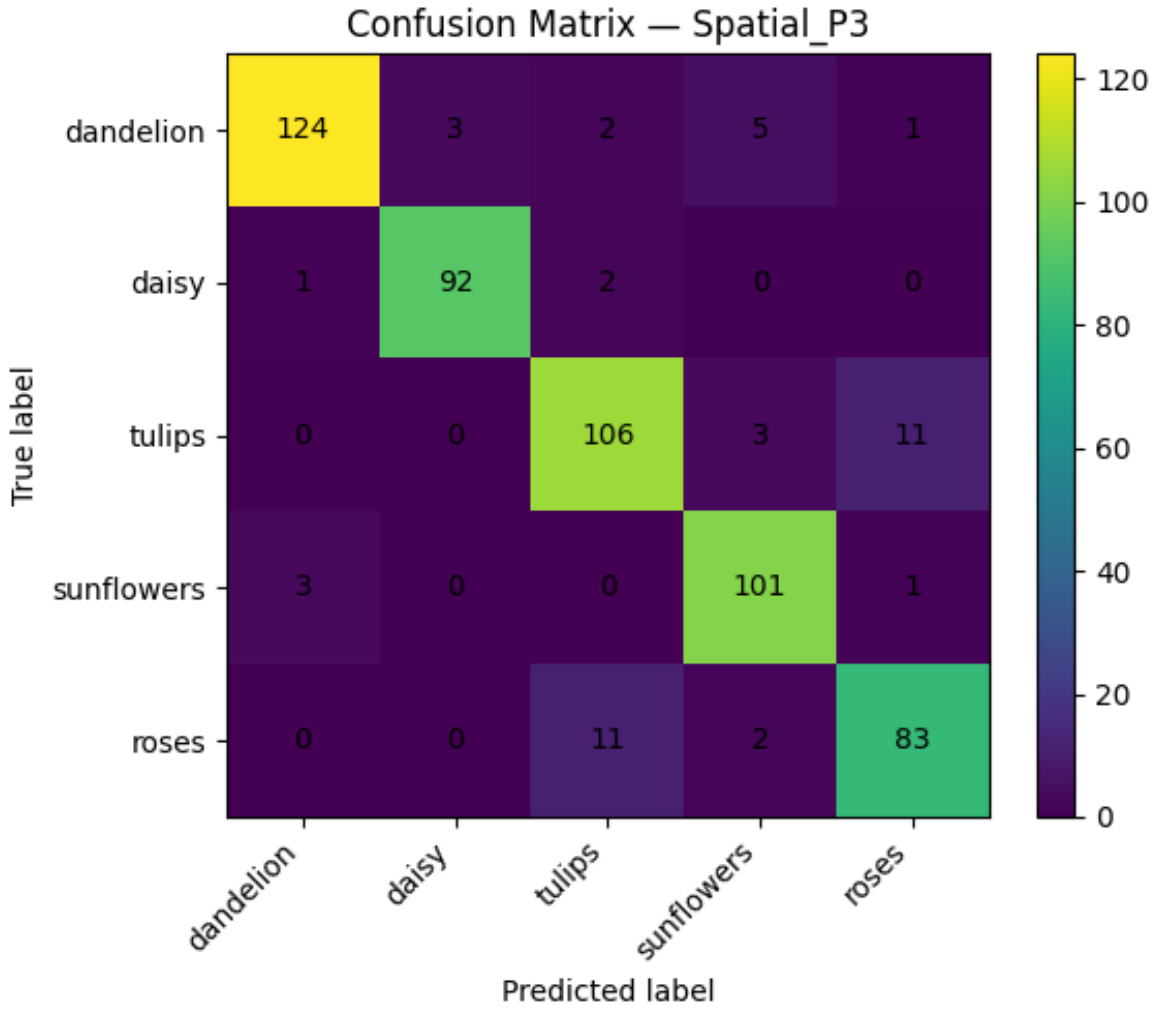


Figure 11: Confusion matrix for the spatial-attention P3 experiment.

Observation: The model achieves 91.83% test accuracy and 0.9182 macro-F1. The class-wise counts in the matrix show which flower categories are most often confused.

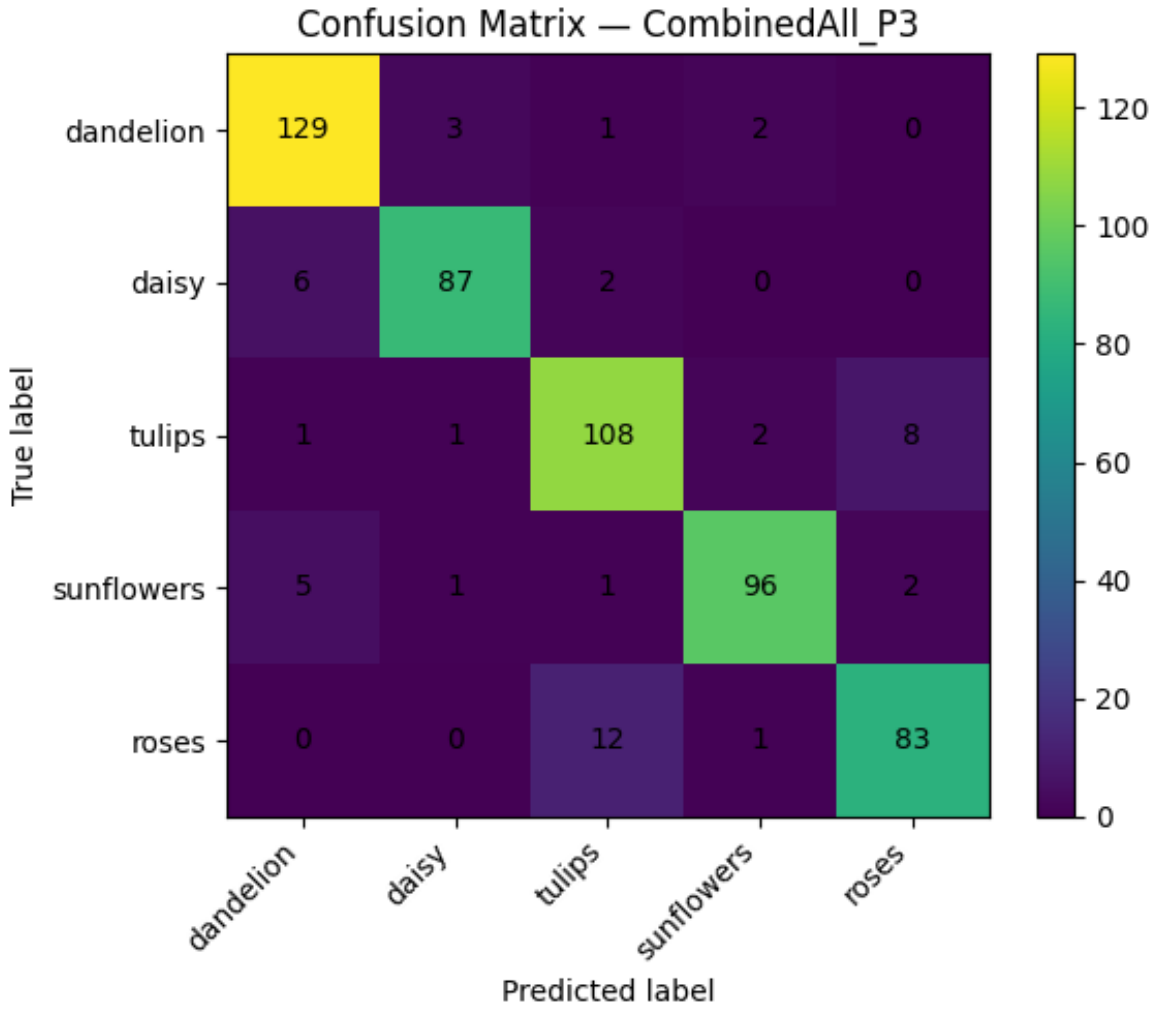


Figure 12: Confusion matrix for CombinedAll_P3.

Observation: CombinedAll_P3 achieves 91.29% test accuracy, below ChannelSpatial_P3 and SelfAttention_P3. Adding multiple attention components together did not guarantee better classification.

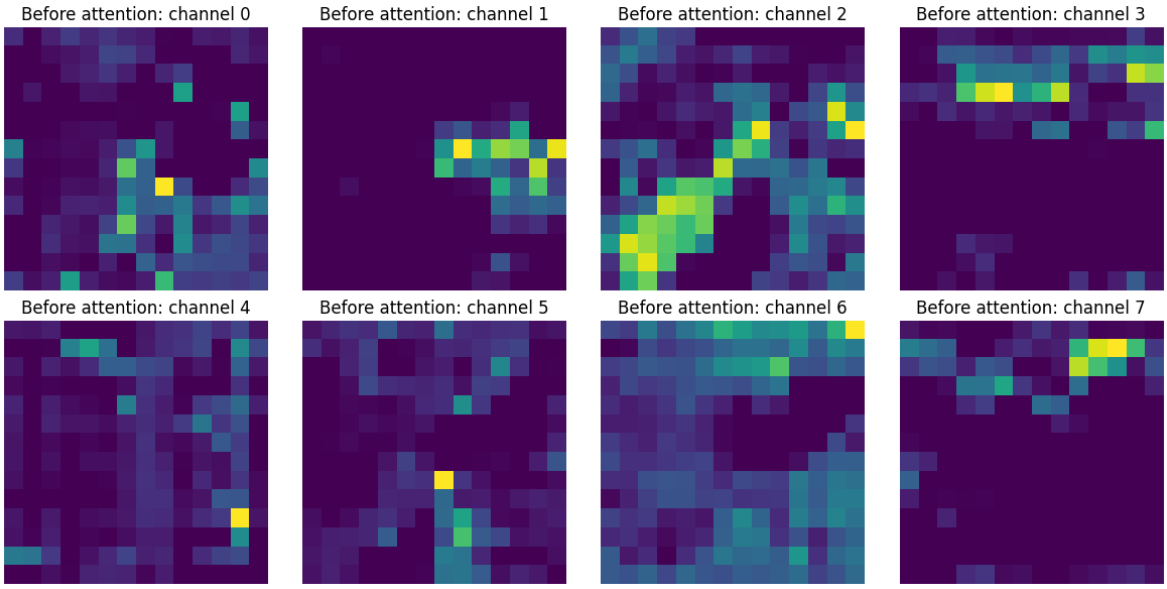


Figure 13: Feature maps before applying attention.

Observation: The feature maps contain activations from different channels of the ResNet50 representation. Individual channels respond to different patterns, textures, edges, and flower structures.

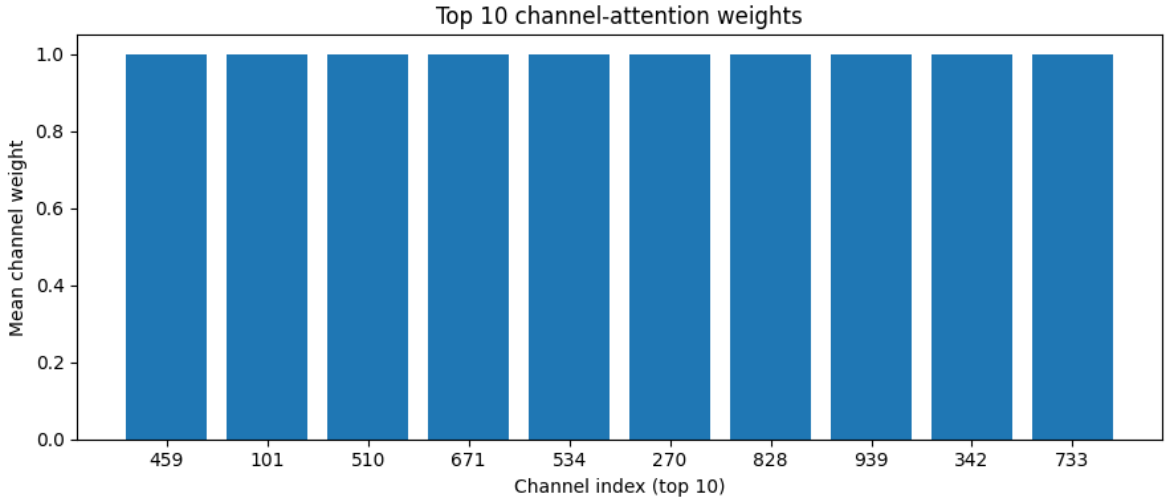


Figure 14: Learned channel-attention weights.

Observation: Channel attention assigns different weights to feature channels, increasing or reducing their contribution to the representation. The weights illustrate the channels emphasized by the module for this example.

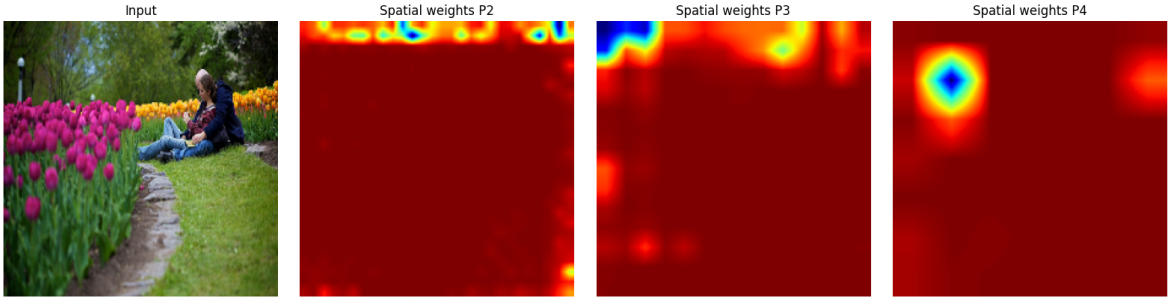


Figure 15: Spatial-attention weight heatmaps.

Observation: The heatmaps indicate which spatial locations receive stronger learned attention. Higher responses show regions emphasized by the attention module, rather than a guaranteed explanation of the prediction.

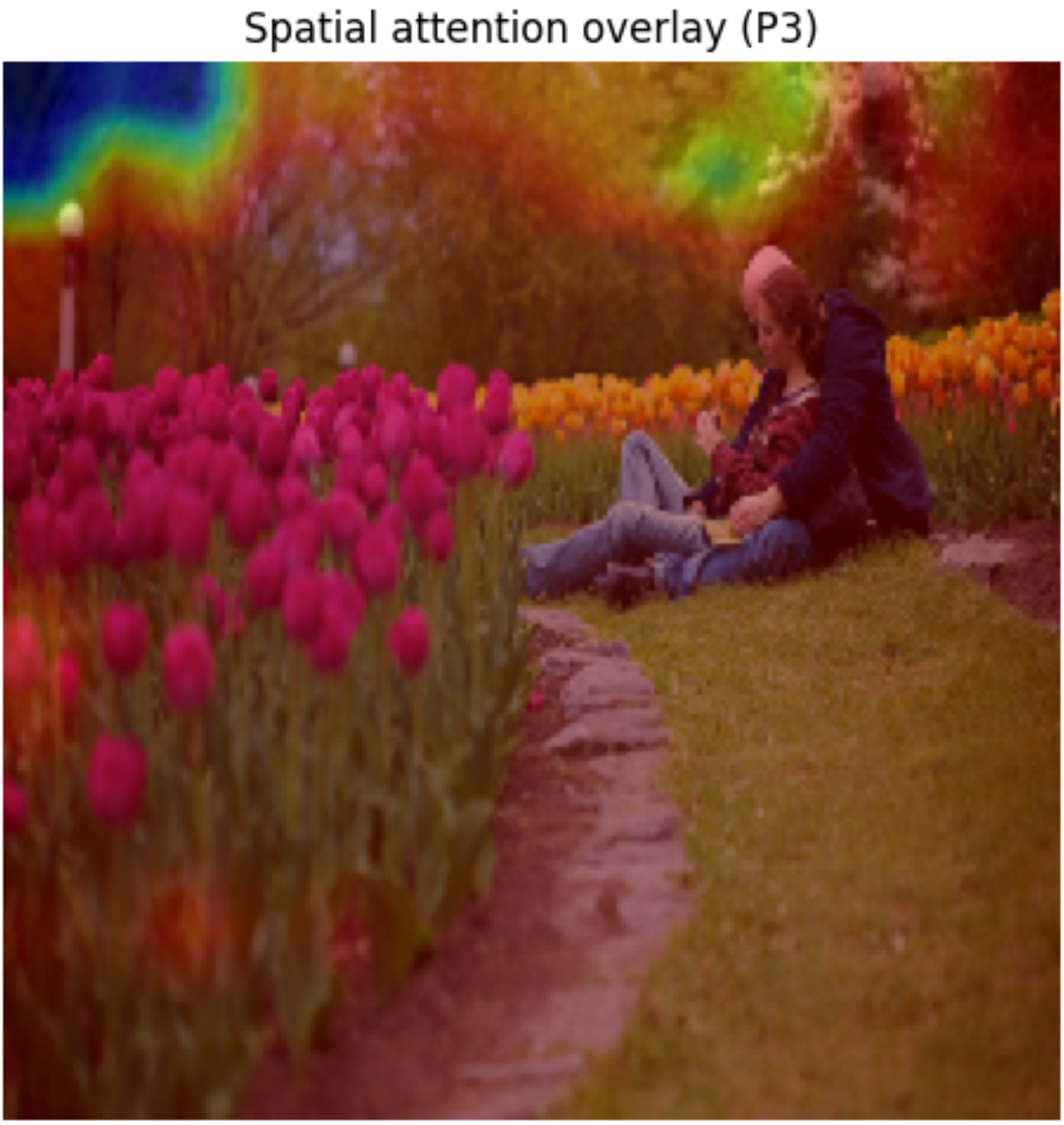


Figure 16: Spatial-attention heatmap overlaid on the input image at P3.

Observation: The overlay makes the emphasized image regions easier to compare with the original flower. It provides a qualitative view of the attention response for this example.



Figure 17: Self-attention response visualization.

Observation: Self-attention models relationships between spatial positions in the feature representation. SelfAttention_P3 obtains the highest test accuracy (93.65%) and macro-F1 (0.9356) in the notebook comparison.

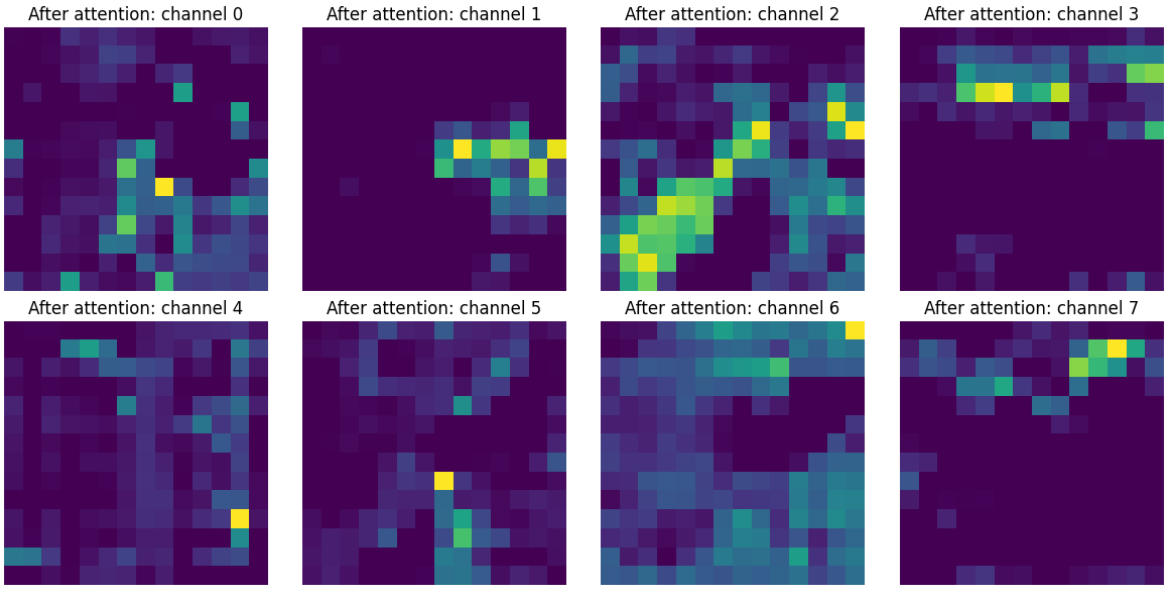


Figure 18: Feature maps after applying attention.

Observation: The post-attention maps show the representation after feature reweighting. Comparing these maps with the pre-attention maps helps illustrate how attention modifies the information passed to later layers.

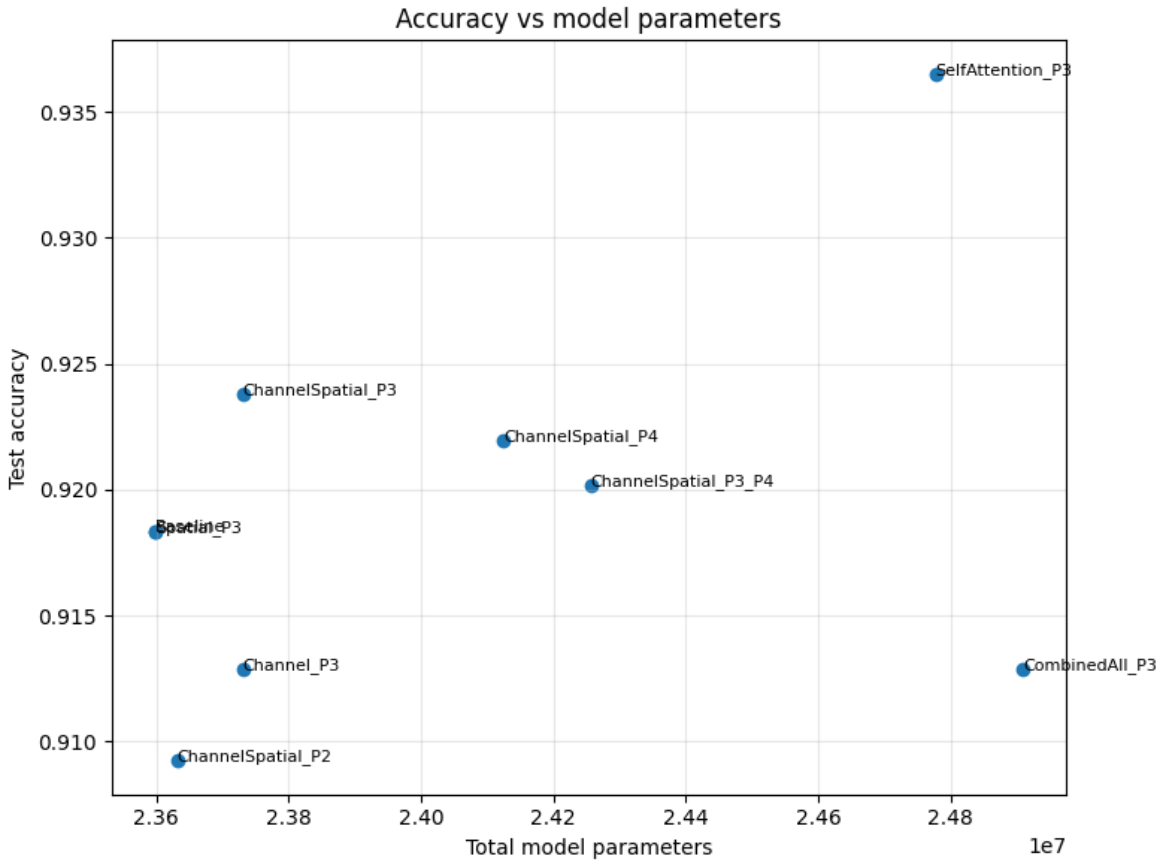


Figure 19: Comparison of test accuracy and model parameter count.

Observation: SelfAttention_P3 gives the highest test accuracy (93.65%) but also has the largest number of trainable parameters among the tested models. ChannelSpatial_P3 offers a smaller accuracy gain with fewer trainable parameters.

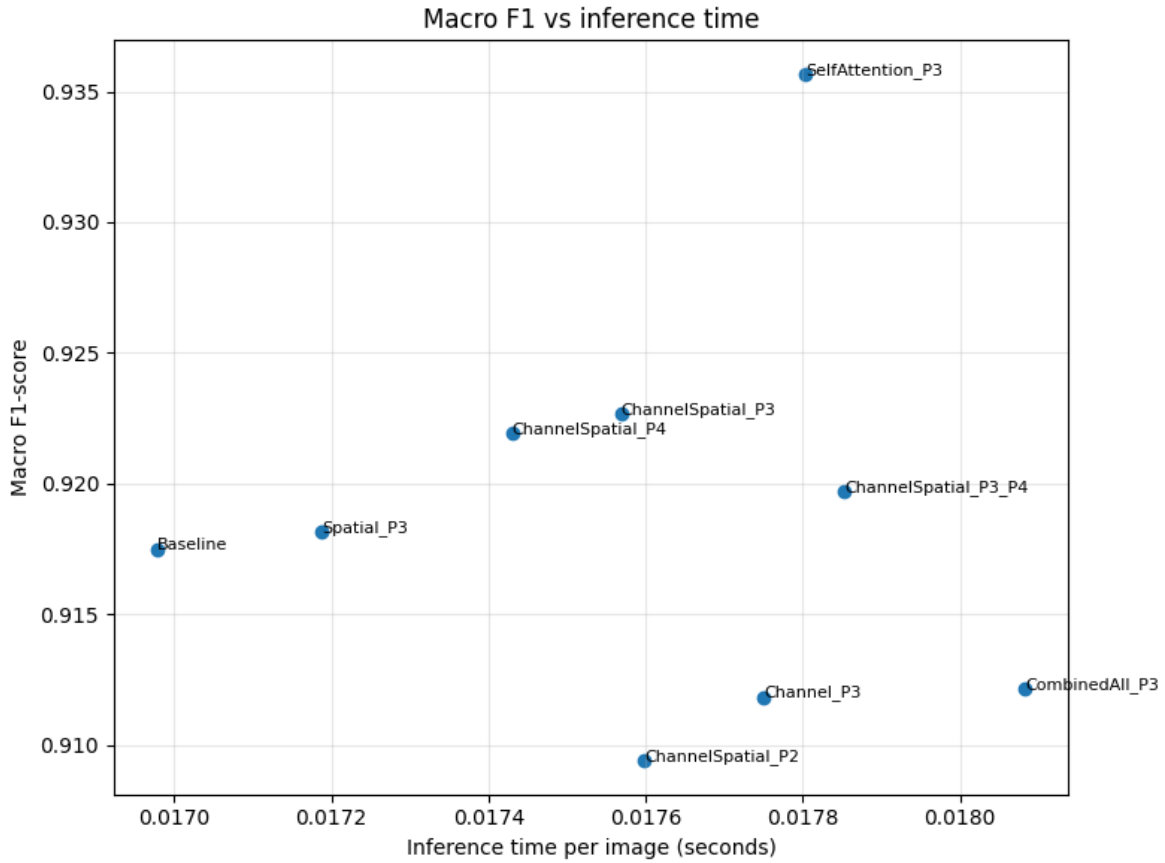


Figure 20: Comparison of macro-F1 score and inference time.

Observation: SelfAttention_P3 achieves the highest macro-F1 score (0.9356). Its measured inference time is about 0.0176 seconds per image, so its accuracy gain comes with a small timing cost relative to the baseline.

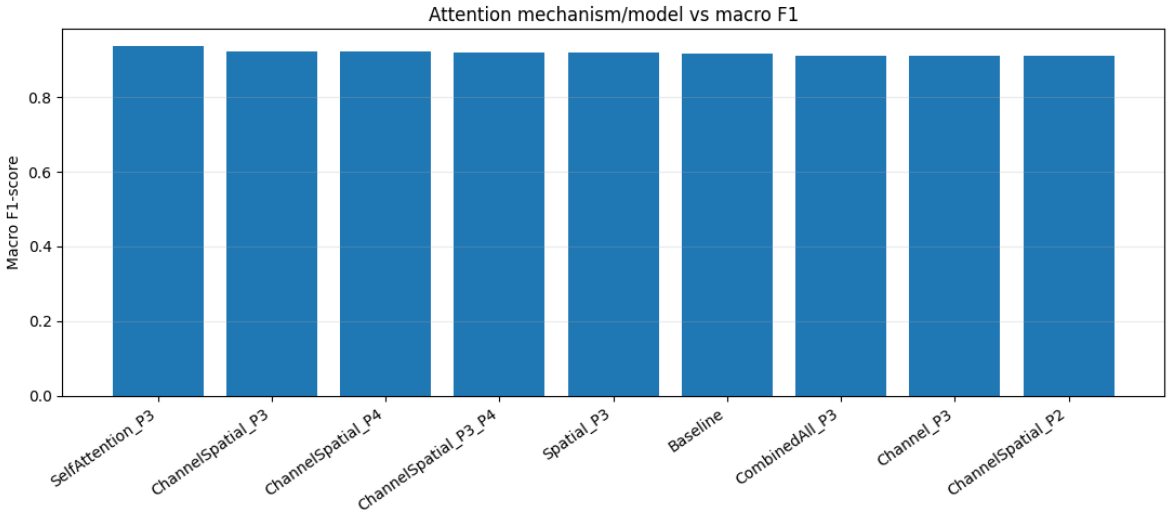


Figure 21: Macro-F1 comparison across models.

Observation: SelfAttention_P3 ranks first with a macro-F1 score of 0.9356, followed by ChannelSpatial_P3 at 0.9227. This indicates the strongest balanced class performance among the evaluated models.

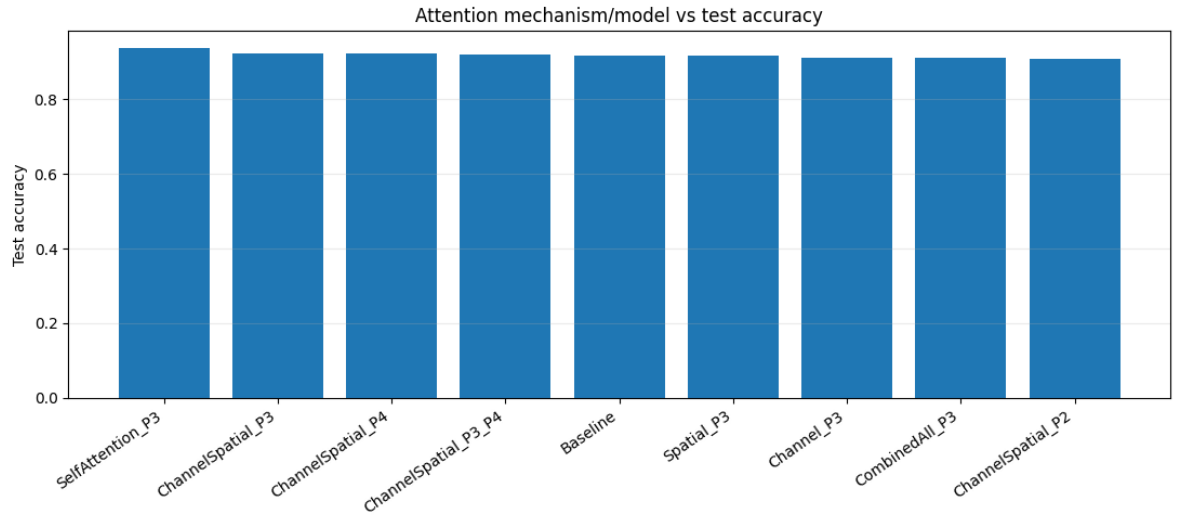


Figure 22: Test accuracy comparison across models.

Observation: SelfAttention_P3 achieves the best test accuracy at 93.65%, followed by ChannelSpatial_P3 at 92.38%. The baseline reaches 91.83%, showing that the benefit depends on the attention design and placement.



Figure 23: Examples of correct and incorrect predictions.

Observation: Correct predictions show examples the model classifies successfully, while incorrect cases reveal remaining classification errors. These examples support qualitative error analysis alongside the aggregate metrics.

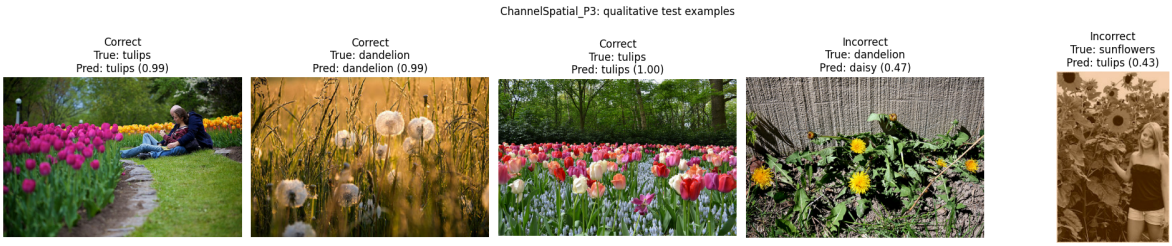


Figure 24: Additional correct and incorrect prediction examples.

Observation: The examples allow visual inspection of cases that may be ambiguous or visually similar across flower classes. Individual examples should be interpreted together with the test-set metrics.



Figure 25: Example test image used for prediction visualization.

Observation: The displayed example provides a qualitative check of the model's prediction. Compare the predicted class with the true label shown by the notebook to determine whether this example was classified correctly.

1.1 Attention at P2, P3, and P4

The results show that ChannelSpatial_P3 (92.38%) performs slightly better than ChannelSpatial_P4 (92.20%) and ChannelSpatial_P2 (90.93%). Therefore, P3 is the strongest of these three tested placements in this run.

32. Recommended Visualization Layout

For one representative test image, students are encouraged to generate a figure in the following format:

Input	Baseline Grad-CAM	Channel Grad-CAM	Spatial Attention Map	Channel+Spatial Grad-CAM

For students completing the additional attention comparison, the figure may be extended to:

Input|Baseline|SE|ECA|CBAM|Self-Attention

using appropriate visualization methods.

33. Required Experimental Analysis

The final report should contain three forms of analysis.

33.1 Quantitative Analysis

Report:

$$\boxed{\text{Accuracy} + \text{Precision} + \text{Recall} + \text{Macro F1}}$$

33.2 Complexity Analysis

Report:

$$\boxed{\text{Parameters} + \text{Inference Time}}$$

33.3 Qualitative Analysis

Report:

$$\boxed{\text{Grad-CAM} + \text{Attention Maps} + \text{Visual Interpretation}}$$

Therefore, the overall evaluation should be:

$$\boxed{\text{Quantitative} + \text{Computational} + \text{Qualitative}}$$

34. Experimental Protocol

To ensure fair comparison, all ablation experiments must use:

- identical training images;
- identical validation images;
- identical testing images;
- identical random seed where possible;
- identical image preprocessing;
- identical augmentation;
- identical backbone;
- identical classifier;
- identical optimizer;
- identical batch size;
- identical number of epochs;
- identical learning-rate schedule;
- identical fine-tuning procedure; and
- identical evaluation metrics.

Only the experimental factor under investigation should change.

For example, when studying attention position:

$$\boxed{\text{Only Attention Position Should Change}}$$

and when studying attention type:

$$\boxed{\text{Only Attention Type Should Change.}}$$

35. Minimum Mandatory Ablation Set

If laboratory time is limited, every student must complete at least the following experiments:

1. Baseline pretrained ResNet50.
2. Channel attention at P_3.
3. Spatial attention at P_3.
4. Channel+Spatial attention at P_3.
5. Channel+Spatial attention at P_2.
6. Channel+Spatial attention at P_4.
7. Self-attention at P_3.
8. Grad-CAM comparison between baseline ResNet50 and one selected attention model.
9. Spatial-attention visualization at P_2, P_3, and P_4.
10. Graphical comparison of the classification performance of the attention models.

This minimum set allows students to study:

Attention Type

and

Attention Position.

36. Additional Student Exploration

Students interested in extended experimentation should select at least two mechanisms from:

SE, ECA, CBAM, Coordinate Attention, Self-Attention

and integrate them with the same ResNet50 backbone.

Students should maintain the same:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;
- epochs;
- fine-tuning strategy; and
- attention position.

The additional investigation should include both:

Quantitative Comparison

and

Qualitative Comparison.

37. Questions Students Must Answer

1. **What is the purpose of residual connections in ResNet50?** Residual connections let information and gradients pass through shortcut paths instead of relying only on the convolution layers. This helps a deep network train more reliably.
2. **What does conv2_x represent compared with conv5_x?** conv2_x is an earlier stage that keeps more spatial detail and learns simpler patterns. conv5_x is much deeper, with smaller feature maps that represent more complex, class-related information.
3. **Why does the spatial resolution decrease with network depth?** The network uses downsampling to reduce the height and width of feature maps. This saves computation and helps later layers combine information from larger areas of the image.
4. **Why does the number of feature channels increase with network depth?** Deeper layers need to represent a wider variety of patterns and object parts. Increasing the channel count gives the network more feature types to work with.
5. **What is meant by low-level, intermediate, and high-level features?** Low-level features include edges, colors, and simple textures. Intermediate features represent patterns or parts, while high-level features capture more meaningful, class-related shapes.
6. **What is the purpose of an attention mechanism?** Attention helps a model emphasize useful information and reduce the influence of less useful features. It changes how features contribute to the final prediction.
7. **What does channel attention attempt to learn?** Channel attention learns which feature channels are more useful for the current input. It assigns weights to channels so important feature types contribute more strongly.
8. **What does spatial attention attempt to learn?** Spatial attention learns which locations in a feature map deserve more emphasis. It produces a spatial weight map that highlights potentially useful regions.
9. **How is channel attention different from spatial attention?** Channel attention weights feature channels, while spatial attention weights positions in the feature map. One asks 'which features?' and the other asks 'where?'.
10. **Why can channel and spatial attention be used sequentially?** Channel attention can first select useful feature types, and spatial attention can then emphasize useful locations in those features. Since the order changes what the second module receives, the results may differ.

11. **Why might channel attention behave differently at conv3_x and conv5_x?** The channels at conv3_x often describe textures and object parts, while conv5_x channels are more semantic. Therefore, the same attention module may assign different importance at these stages.
12. **Why can spatial attention be useful at an intermediate ResNet stage?** An intermediate stage can preserve more location detail than the deepest stage while already detecting meaningful patterns. Spatial attention can use this balance to highlight relevant image regions.
13. **Why is self-attention computationally expensive when the feature map has high spatial resolution?** Self-attention compares every spatial position with every other position. With many positions, these comparisons require substantial memory and computation.
14. **What happens to the size of the self-attention matrix when the spatial resolution increases?** If there are $N = H \times W$ positions, the matrix contains $N \times N$ entries. Doubling both height and width makes four times as many positions and sixteen times as many matrix entries.
15. **How does attention performance vary when the insertion position is changed?** The result can change because each ResNet stage contains features at a different level of detail. In this notebook, ChannelSpatial_P3 reached 92.38% accuracy, compared with 90.93% at P2 and 92.20% at P4.
16. **How does attention at P_2 differ visually from attention at P_4?** P2 usually preserves finer spatial detail, so its attention can look more localized. P4 has lower spatial resolution and more semantic features, so its attention map is generally coarser; confirm the pattern in the plotted maps.
17. **Does applying attention at multiple stages always improve classification performance?** No, adding more attention blocks does not guarantee better accuracy. In this run, ChannelSpatial_P3P4 achieved 92.01%, below the 92.38% obtained by ChannelSpatial_P3 alone.
18. **Does Channel→Spatial attention behave differently from Spatial→Channel attention?** Yes, the order can change the result because the second module receives features already modified by the first. The two orders should be compared using the same split and training settings.
19. **How does the spatial-attention kernel size influence the attention map?** A larger kernel considers a wider neighborhood and may create smoother, broader spatial responses. A smaller kernel focuses on more local patterns, but the best size should be decided from experimental results.
20. **How does SE attention differ from ECA attention?** SE uses a small bottleneck multilayer perceptron to learn channel weights. ECA uses a lightweight one-dimensional convolution across channel descriptors, avoiding the dimensionality-reduction bottleneck.
21. **How does CBAM combine channel and spatial attention?** CBAM applies channel attention and then spatial attention in sequence. This lets the model reweight both the feature channels and the locations within the feature map.
22. **How does self-attention differ from channel attention?** Channel attention learns the importance of feature channels. Self-attention instead models relationships between spatial positions, allowing information from different parts of the feature map to interact.
23. **Does the attention mechanism with the largest number of parameters necessarily provide the highest accuracy?** No, more parameters do not automatically mean better predictions. In this notebook, SelfAttention_P3 achieved the highest accuracy at 93.65%, but model size and inference cost still matter.
24. **What is the difference between an internal attention map and Grad-CAM?** An internal attention map is produced by an attention module as part of the model. Grad-CAM uses gradients for a selected class score to create a post-hoc heatmap of influential image regions.
25. **Why is Grad-CAM called a post-hoc explanation technique?** Grad-CAM is generated after a model has made a prediction, using gradients and feature maps to inspect that prediction. It does not itself change how the model was trained.
26. **Does the attention model suppress irrelevant background regions?** It may reduce the weight of some background locations, but this should not be assumed for every image. The overlays need to be inspected to see which areas are actually emphasized or deemphasized.
27. **Are the attention maps of correctly classified and incorrectly classified images different?** They can be different, but this needs to be checked across several examples. Comparing both groups may reveal whether errors occur when attention focuses on background or ambiguous flower features.
28. **Which flower classes benefit most from attention?** This cannot be determined from overall accuracy alone. Compare per-class precision, recall, F1-scores, and confusion matrices for the baseline and attention models before naming the classes.
29. **What information can be obtained from a confusion matrix that cannot be obtained from overall accuracy alone?** A confusion matrix shows which classes are predicted correctly and which are confused with one another. Overall accuracy gives only one summary number and hides these class-specific errors.
30. **Why is macro F1 useful in addition to accuracy?** Macro F1 calculates an F1-score for each class and averages them equally. It helps reveal weak performance on a class even when the overall accuracy looks high.
31. **What is meant by the accuracy–complexity trade-off?** It means judging whether an accuracy improvement is worth the extra parameters, memory, or computation. A slightly less accurate model may still be preferable if it is much lighter or faster.

32. **What is meant by the F1–inference-time trade-off?** A model may improve macro F1 but take longer to produce predictions. The preferred model depends on whether balanced classification performance or prediction speed matters more for the task.
33. **Why must every ablation study use the same dataset split?** Using the same training, validation, and test examples makes the comparison fair. Otherwise, a model might appear better simply because it received an easier split.
34. **Why must the same training protocol be maintained when comparing different attention mechanisms?** Keeping the optimizer, epochs, data processing, and other settings consistent reduces unrelated differences. This makes it more reasonable to attribute performance changes to the attention design.
35. **What conclusions can be drawn from the Grad-CAM comparison?** Grad-CAM can show whether the baseline and attention model highlight similar or different image regions. The heatmaps are qualitative evidence and should not be treated alone as proof of the model's reasoning.

38. Observations to be Recorded

Students should record the following during experimentation:

1. Input and output feature dimensions for each attention module.
2. Training accuracy.
3. Validation accuracy.
4. Testing accuracy.
5. Training loss.
6. Validation loss.
7. Macro precision.
8. Macro recall.
9. Macro F1-score.
10. Number of trainable parameters.
11. Total number of model parameters.
12. Average inference time.
13. Confusion matrix.
14. Channel-attention weights.
15. Spatial-attention maps.
16. Self-attention responses.
17. Grad-CAM maps.
18. Attention maps for different insertion positions.

39. Suggested Laboratory Report Structure

The laboratory report may be organized as follows:

1. **Aim**
2. **Dataset Description**
3. **ResNet50 Architecture**
4. **Attention Mechanisms**
5. **Proposed Experimental Architecture**
6. **Experimental Setup**
7. **Attention-Type Ablation**
8. **Attention-Position Ablation**
9. **Performance Results**
10. **Complexity Analysis**
11. **Training and Validation Graphs**
12. **Confusion Matrix Analysis**
13. **Grad-CAM Analysis**
14. **Internal Attention-Map Analysis**
15. **Qualitative Comparison**
16. **Additional Attention-Mechanism Comparison**
17. **Discussion**
18. **Conclusion**

40. Discussion Guidelines

Students should not merely reproduce numerical results.

The discussion should explain:

- why a particular attention mechanism produced a change in performance;
- how attention location influences feature refinement;
- how feature resolution changes attention behavior;
- how model complexity changes with different mechanisms;
- whether performance improvements justify additional computational cost;
- whether attention maps correspond to meaningful object regions;
- whether attention helps suppress the background;
- whether attention behaves differently for incorrect predictions; and
- whether quantitative and qualitative observations support each other.

Students should avoid making conclusions based on a single image.

Qualitative observations should preferably be based on several representative test images.

41. Expected Outcome

At the end of the experiment, students should understand that an attention mechanism is not an isolated module whose behavior is independent of the backbone.

Its effectiveness depends on:

$$\boxed{\begin{aligned}\text{Attention Effectiveness} = {} & \text{Attention Type} \\ & + \text{Attention Position} \\ & + \text{Feature Resolution} \\ & + \text{Feature Semantics} \\ & + \text{Model Complexity.}\end{aligned}}$$

Students should also understand that classification performance alone does not provide a complete explanation of an attention mechanism.

A comprehensive analysis should therefore combine:

$$\boxed{\text{Classification Performance}}$$

with

$$\boxed{\text{Computational Complexity}}$$

and

$$\boxed{\text{Qualitative Visualization.}}$$

Therefore:

$$\boxed{\text{Complete Attention Analysis} = \text{Quantitative} + \text{Computational} + \text{Qualitative}}$$

The final conclusions must be based on the student's experimental results rather than on assumptions about which attention mechanism should perform best.

42. References

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8. TensorFlow Documentation: https://www.tensorflow.org/
9. Keras Documentation: https://keras.io/
10. TensorFlow Datasets: https://www.tensorflow.org/datasets

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2 Project Repository and Notebook Links

The source code and experimental notebook for this project are provided below for reference and reproducibility.

- **GitHub Repository:** Click here to access the project source code
- **Colab Notebook:** Click here to access the experimental notebook

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